Digital Twin for Type 2 Diabetes

Predicting Glucose-Spike Risk Up to 2 Hours Ahead

Team TYST | St. Joseph's College of Engineering

Digital Twin Challenge 2026 - Prototype & Code Submission

The Problem

- Type 2 diabetes requires continuous awareness of changing glucose levels.
- Traditional monitoring is mostly reactive: abnormal glucose is identified after it occurs.
- Glucose patterns are influenced by activity, sleep, heart rate, lifestyle and patient history.
- Need: an early-warning system that combines historical and real-time patient context.

Our Solution

- A patient-specific Digital Twin that fuses EHR and wearable/time-series data.
- Static EHR: age, sex, BMI, HbA1c and years since diagnosis.
- Dynamic data: glucose, heart rate, steps and sleep.
- Output: Low / Moderate / High risk of a glucose spike within the next 2 hours.
- Designed as clinical decision support - not a replacement for a doctor.

System Architecture

- Patient data enters through two streams: EHR + wearable/sensor data.
- Preprocessing cleans and aligns the time-series.
- Feature engineering creates glucose trends, rolling statistics, activity and time-of-day features.
- Static and dynamic features are fused and passed to the ML model.
- Prediction and explanations are displayed in the Streamlit doctor dashboard.

Dataset & Privacy

- Synthetic dataset generated for the proof-of-concept.
- 50 patients x 14 days at 5-minute intervals (~200,000 rows).
- Dynamic variables: glucose, heart rate, steps and sleep.
- Static variables: age, sex, BMI, HbA1c and years since diagnosis.
- Target: whether glucose exceeds 180 mg/dL within the next 2 hours.
- No personally identifiable patient information is used.

AI / ML Pipeline

- Binary classification for 2-hour glucose-spike risk.
- Features use past data only to reduce data leakage.
- Models: baseline rule, Logistic Regression, and tree-based model (Random Forest / Gradient Boosting).
- Patient-level train/validation/test split prevents the same patient appearing across sets.
- Evaluation: ROC-AUC, PR-AUC, recall, precision, F1 and confusion matrix.
- Risk probability is mapped to Low / Moderate / High.

Doctor Dashboard

- Patient selector and EHR profile.
- Current glucose, heart rate, steps and sleep metrics.
- Glucose trend with the 180 mg/dL threshold.
- 2-hour risk gauge: Low / Moderate / High.
- 'Why this risk?' explanation using feature importance.
- What-if analysis for lifestyle changes such as steps and sleep.

Prototype Outcome

- End-to-end pipeline implemented: data generation -> feature engineering -> model -> prediction -> dashboard.
- Live Streamlit dashboard is deployed.
- Public GitHub repository contains source code, model workflow, documentation and architecture diagram.
- Model metric fields should be updated with actual training output before final submission.
- Current results are proof-of-concept results on synthetic data and are not clinical-performance claims.

Limitations & Future Work

- Synthetic data cannot represent the full variability of real patients.
- No clinical validation or regulatory review has been performed.
- Future: validate with consented, de-identified CGM + EHR datasets.
- Add medication and meal logging.
- Develop per-patient personalization and clinician-validated alert thresholds.

Conclusion

- Proactive rather than reactive diabetes monitoring.
- Digital Twin combines patient history with continuously changing physiological data.
- 2-hour risk prediction can support earlier clinical attention and personalized conversations.
- Prototype demonstrates a practical foundation for future real-world validation.
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